

Passivity-Based Stabilization of *LCL*-Type Grid-Connected Inverters via a General Admittance Model

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Abstract—For *LCL*-type grid-connected inverters, inverter-side current, grid-side current, capacitor current and voltage are four commonly used variables in the control serving as different roles, such as control objective, active damping, synchronization, etc. This article builds a general admittance model, in which the total output admittance is decomposed into four paralleled subadmittance corresponding to the above four variables in the control law, to facilitate the passivity-based stability assessment and controller parameter optimization for different control objectives and schemes. Based on the passivity analysis, an appropriate combination scheme of subadmittance and a simple unified analytical controller parameter design method are proposed to obtain a passive output admittance from 0 Hz up to Nyquist frequency for either inverter-side or grid-side current control. Therefore, it can not only ensure the stable operation of *LCL*-type inverters regardless of grid impedance (either inductive or capacitive) but also avoid the conventional design constraints of an *LCL* filter, i.e., $f_r < \text{or} > f_s/6$ is required for the single-loop inverter-side or the grid-side current control. Finally, experiments verify the theoretical results.

Index Terms—Inverter control, grid-connected inverter, *LCL* filter, passivity, stability.

I. INTRODUCTION

WITH the development of highly penetrated renewable energy generation systems, grid-connected inverters have been extensively employed as flexible and efficient grid interfaces in the modern power system [1]. *LCL* filters have been commonly adopted to filter out switching harmonics generated by inverters. Compared with *L* filters, *LCL* filters have better harmonic attenuating ability and allow the use of low inductance

inductors, resulting in a cost-effective solution [2]. Unfortunately, the inherent resonance hazard of *LCL* filters has the tendency to destabilize the inverter systems. Stabilization of the resonance of *LCL* filters is a challenging task and has attracted significant attention [3]–[24].

Continuous research efforts have been made to propose solutions to cope with the resonance of *LCL* filters, among which the most direct way is to put a damping resistor in series with the filter capacitor, named as passive damping method [2]. This method is very simple and highly reliable but results in high power losses and degradation of the harmonics attenuation ability [3]. A more attracting way is to use the active damping method, the principle of which is to emulate a virtual resistor by feeding back variable as the damping terms, e.g., filter capacitor current [4]–[6], filter capacitor voltage [7], [8], inverter-side current [9], or grid-side current [10], [11].

In a digitally controlled system, the computation and pulsewidth modulation (PWM) delays have significant effects on the active damping performances. Dannehl *et al.* [12], [13] indicate that the effectiveness of active damping is closely related to the ratio of sampling frequency (f_s) to the *LCL* resonance frequency (f_r), but the nature of this relationship is not known. With a typical delay of one and a half sampling periods ($1.5T_s$), Parker *et al.* [14] further figure out one-sixth of the sampling frequency ($f_s/6$) as the critical resonance frequency (f_{crit}). The stable region of f_r for the single-loop grid-side current control is the interval of (f_{crit} , $f_s/2$) [15] while the one for the single-loop inverter-side current control is the interval of (0, f_{crit}) [16]. Since the inverter-side current is the superposition of the grid-side and capacitor currents [9], the same stable region is derived for the capacitor current feedback active damping (CCF-AD) in [17]. This imposes a constraint on designing the *LCL* filter, i.e., its resonance frequency ought to be within the stable region [25], to avoid instability. However, the unknown variation of grid impedance still may shift the filter resonance frequency to the unstable region. To mitigate such possibility, one way is to widen the stable region, which for either single-loop inverter-side current control or CCF-AD, is to increase f_{crit} by compensating time delay in the control loop [17]–[22]. For the single-loop grid-side current control, the reverse (decreasing f_{crit}) must be performed by adding a proper time delay [16]. However, it will undesirably shrink its bandwidth, and hence worsen its

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dynamics. With the same delay of $1.5T_s$, f_{crit} for the capacitor voltage feedforward active damping (CVF-AD) is derived via the virtual impedance model, i.e., $f_s/3$ [23]. Li and Wu [24] lift the f_{crit} approaching Nyquist frequency via the full delay compensation in the CVF-AD loop. Nevertheless, active damping methods are usually designed on the basis of the classical closed-loop transfer functions and under the assumption of inductive grid condition, which is not applicable in the capacitive grid or the system that comprises multiple inverters [26], [27]. As Akhavan *et al.* [22] demonstrate, although the stable region of CCF-AD is expanded up to Nyquist frequency, the system is still unstable under specific grid conditions.

The frequency-domain passivity theory offers another effective and attractive way to assess the stability of grid-connected inverters [28]. According to the passivity theory, if critical grid resonances fall within regions where the output admittance of the inverter acts passively, i.e., has a nonnegative real part, then the destabilization is generally prevented [28]. The passivity characteristic of the inverter-side current-controlled *LCL*-filtered inverter has been analyzed in [29] and [30]. The frequency region for the inverter output admittance having a negative real part is identified. The damping control based on CVF-AD is also developed to lift the passive region up to the Nyquist frequency [30]. However, an extra analog circuit-based high-pass filter is added and increases the cost as well as implementation complexity. The passivity characteristic of the grid-side current-controlled *LCL*-filtered inverter is studied in [27], and a discrete derivative controller-based damping injection method is used to enhance the passivity of the inverter. However, the nonpassive region remains and still may trigger the system resonance. Recently, a bi-quad filter-based delay compensation method is proposed to expand the stable region of CCF-AD up to the Nyquist frequency, plus the point of common coupling (PCC) voltage feedforward scheme, and all frequencies passivity of the output admittance is achieved for the grid-side current-controlled *LCL*-filtered inverter [22]. However, the bi-quad filter will increase the complexity of parameter selection and algorithm implementation, besides the used graphical parameter tuning method is also time consuming.

This article builds, in Section II, a general admittance model for the *LCL*-type grid-connected inverter, in which the total inverter output admittance is decomposed into four paralleled subadmittance corresponding to the four variables in control law i.e., inverter-side current, grid-side current, capacitor current, and capacitor voltage. In Section III, the passivity of different control objectives and schemes are analyzed using the above subadmittance. It is found that the same all frequencies passive output admittance for both inverter-side and grid-side current control can be achieved by an appropriate combination of subadmittance as well as parameter selection. Then, a simple unified analytical controller design method is developed in Section IV, whereby passive output admittance in all frequencies is obtained for both inverter-side and grid-side current control. Time-consuming iterations in the controller design are thus avoided. Experiments are performed in Section V to verify the theoretical results. Section VI summarizes this article.

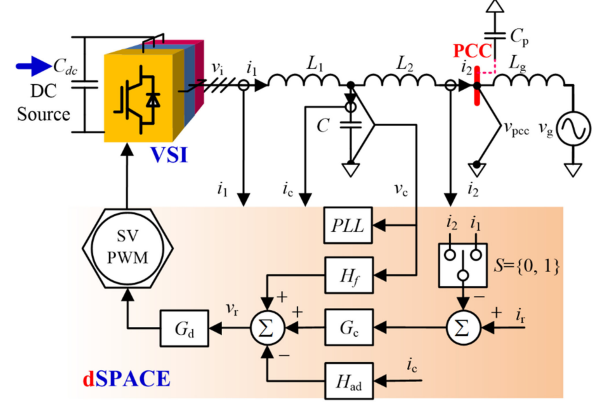


Fig. 1. System configuration of a three-phase *LCL*-type grid-connected inverter.

II. GENERAL ADMITTANCE MODEL OF THE *LCL*-TYPE GRID-CONNECTED INVERTER

A. System Description and Modeling

Fig. 1 shows the system configuration of a three-phase voltage-source inverter feeding into the grid through an *LCL*-type filter. In the figure, L_1 , L_2 , and C are the inverter-side inductor, grid-side inductor, and filter capacitor, respectively. L_g is the grid inductor, and C_p represents the power factor correction (PFC) capacitor connected at PCC. The parasitic resistances of all inductors are neglected for the worst case with zero passive damping.

For the objectives of either inverter-side or grid-side current regulation, both inverter-side current i_1 and grid-side current i_2 are fed back to the current controller through a selector, S . Note that $S = 1$ means the inverter-side current control, while $S = 0$ stands for the grid-side current control. Filter capacitor current i_c and filter capacitor voltage v_c are also fed back to the controller for the sake of CCF-AD and CVF-AD, respectively. v_c is also fed to the phase-locked loop (PLL) to generate the grid synchronization signal. In addition, v_{pcc} and v_g in Fig. 1 are PCC voltage and grid voltage, respectively.

With the symbols as defined in Fig. 1, the dynamic equations of the *LCL*-filtered grid-connected inverter can be expressed as follows [31]:

$$\begin{cases} L_1 \frac{di_1}{dt} = v_i - v_c \\ L_2 \frac{di_2}{dt} = v_c - v_{\text{pcc}} \\ i_1 = i_c + i_2 \\ C \frac{dv_c}{dt} = i_c. \end{cases} \quad (1)$$

Applying the Laplace transform to (1), the corresponding s -domain equations can be obtained as (2), where the capital case symbols represent the corresponding s -domain variables

$$\begin{cases} L_1 s \cdot I_1(s) = V_i(s) - V_c(s) \\ L_2 s \cdot I_2(s) = V_c(s) - V_{\text{pcc}}(s) \\ I_1(s) = I_c(s) + I_2(s) \\ C s \cdot V_c(s) = I_c(s). \end{cases} \quad (2)$$

B. Control Scheme

To build a general model, both inverter-side and grid-side currents are fed back, and then passed through a selector to the current controller, i.e., a phase compensated proportional resonant (PR) controller $G_c(s)$ [19]. In addition, both feedback of the capacitor current through $H_{ad}(s)$ and feedforward of the capacitor voltage through $H_f(s)$ are also included in the control scheme, to give the following inverter voltage reference:

$$V_r(s) = \{I_r(s) - [S \cdot I_1(s) + (1 - S)I_2(s)]\} G_c(s) - H_{ad}(s)I_c(s) + H_f(s)V_c(s) \quad (3)$$

where I_r is the reference current, and $G_c(s)$ is with the form of

$$G_c(s) = K_p + K_I \frac{s \cos \phi_1 - \omega_1 \sin \phi_1}{s^2 + \omega_{rc}s + \omega_1^2} \quad (4)$$

where K_p , K_I , ω_1 , ω_{rc} , and ϕ_1 are the proportional gain, resonant gain, resonant angular frequency, cutoff angular frequency, and compensation angle of the PR controller, respectively.

The inverter output voltage v_i is created by PWM. The dc-link voltage is regarded as constant for simplicity and bandwidth of the PLL is designed smaller than the grid fundamental frequency to avoid unintentional low-frequency interaction [30]. Taking the typical delay of $1.5T_s$ into consideration, the inverter output voltage can be expressed as

$$V_i(s) = V_r(s)G_d(s) \quad (5)$$

where $G_d(s) = e^{-T_d s}$ and $T_d = 1.5T_s$.

C. Proposed General Admittance Model

Substituting (3) and (5) into (2), with i_2 and v_c regarded as the output and a disturbance, respectively, giving

$$I_2(s) = T_c(s)I_r(s) - Y(s)V_c(s) \quad (6)$$

where the inner closed-loop system ($T_c(s)$) and the inner total output admittance ($Y(s)$), respectively, are given by

$$T_c(s) = \frac{G_c(s)G_d(s)}{L_1s + G_c(s)G_d(s)} \quad (7)$$

and

$$Y(s) = \frac{1 + L_1Cs^2 + \{[S \cdot G_c(s) + H_{ad}(s)] \cdot Cs - H_f(s)\}G_d(s)}{L_1s + G_c(s)G_d(s)} \quad (8)$$

The equivalent circuit depicted in Fig. 2 is obtained, in which $Z_2(s) = L_2s$, and $Z_g(s)$ is the grid impedance. The total output admittance $Y(s)$ is further decomposed into four paralleled subadmittance, as shown in Fig. 2(b). The subadmittance correspond to the different control variables used in (3) and give expressions as

$$Y(s) = S \cdot \left[\frac{Y_1(s)}{L_1s + G_c(s)G_d(s)} + Cs \right] + (1 - S)Y_2(s)$$

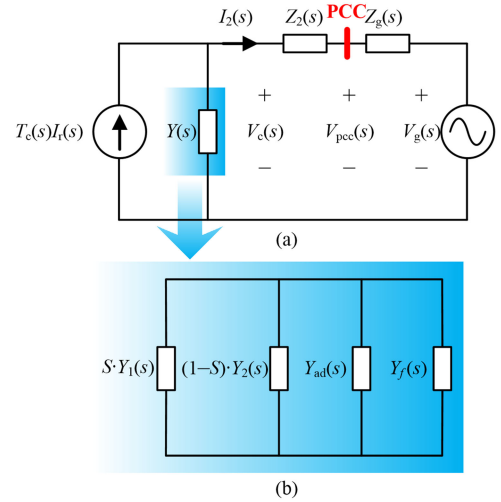


Fig. 2. Equivalent circuit for the grid-connected inverter. (a) Equivalent circuit. (b) Equivalent paralleled subadmittance for the total output admittance.

$$Y_2(s) = \frac{1 + L_1Cs^2}{L_1s + G_c(s)G_d(s)} + \frac{Y_{ad}(s)}{L_1s + G_c(s)G_d(s)} + \frac{Y_f(s)}{L_1s + G_c(s)G_d(s)} \quad (9)$$

where $Y_1(s)$, $Y_2(s)$, $Y_{ad}(s)$, and $Y_f(s)$ represent subadmittance corresponding to the single-loop inverter-side current control, single-loop grid-side current control, CCF-AD, and CVF-AD, respectively.

III. PASSIVITY-BASED STABILITY ANALYSIS

A. Overall Closed-Loop System Stability Condition

According to in Fig. 2(a), combining (6) with $V_c(s) = V_g(s) + I_2(s)[Z_2(s) + Z_g(s)]$, the overall closed-loop current response seen from the v_g , which is the stiff grid voltage behind the grid impedance, can be derived as

$$I_2(s) = \frac{T_c(s)}{1 + [Z_2(s) + Z_g(s)]Y(s)} I_r(s) - \frac{Y(s)}{1 + [Z_2(s) + Z_g(s)]Y(s)} V_g(s). \quad (10)$$

From (10), it can be observed that the overall closed-loop system is stable if the following two conditions hold [27], [29], [30].

- 1) The inner closed-loop system ($T_c(s)$) is asymptotically stable.
- 2) The roots of $1 + [Z_2(s) + Z_g(s)]Y(s) = 0$ are all located in the left half of the s -plane.

In the derived system model, both inverter-side and grid-side current control share the same inner closed-loop system, i.e., a first-order system with a pure delay element. Therefore, the first stable condition is easy to achieve [29]. Then, the second stability condition can be analyzed by applying the Nyquist criterion to

the open-loop transfer function $[Z_2(s) + Z_g(s)]Y(s)$. However, if $[Z_2(s) + Z_g(s)]$ and $Y(s)$ are both passive, then the second stability condition is guaranteed [29]. As long as $[Z_2(s) + Z_g(s)]$ represents a resistive-inductive-capacitive (RLC) network, it is obviously passive. Remaining is, thus, to make $Y(s)$ passive.

B. Single-Loop Inverter-Side Current Control

The stable region of the single-loop inverter-side current control is first identified via the passivity characteristics of the corresponding subadmittance given in (9). The resonant part of the current controller is neglected because of its negligible effect outside the fundamental frequency [18], [22]. Hence, the current controller can be simplified as

$$G_c(s) = \omega_c L_1 \quad (11)$$

where ω_c is the crossover angular frequency of the inner current control loop.

Substituting (11) into (9), and taking the real part of $Y_1(s)$ yields

$$\text{Re}\{Y_1(j\omega)\} = \frac{\omega_c \cos(\omega T_d)}{[\omega_c^2 + \omega^2 - 2\omega_c \omega \sin(\omega T_d)]L_1}. \quad (12)$$

As the denominator of (12) is always positive, the passivity of $Y_1(s)$ can be investigated according to the numerator of (12). It can be easily concluded that $\text{Re}\{Y_1(j\omega)\}$ changes sign from positive to negative at $\omega = \pi/(2 \times T_d) = \omega_s/6$. This is intuitively illustrated in Fig. 3(a), where normalization by $\omega_s L_1$ is made to obtain dimensionless values. $f_s/6$ can be regarded as the critical resonant frequency, in agreement with the results found in [16]. If the system resonant frequency (f_r), which is defined as (13), located in the passive region of $Y_1(s)$, i.e., $f_r < f_s/6$, the system is stable, and vice versa. This is confirmed by the Nyquist curves in Fig. 3(b)

$$f_r = \frac{1}{2\pi} \sqrt{\frac{L_1 + L_2 + L_g}{L_1(L_2 + L_g)C}}. \quad (13)$$

C. Single-Loop Grid-Side Current Control

For the single-loop grid-side current control, taking the real part of $Y_2(s)$ yields

$$\text{Re}\{Y_2(j\omega)\} = \frac{(1 - L_1 C \omega^2) \omega_c \cos(\omega T_d)}{[\omega_c^2 + \omega^2 - 2\omega_c \omega \sin(\omega T_d)]L_1}. \quad (14)$$

From (14), it can be concluded that $\text{Re}\{Y_2(j\omega)\}$ is negative in the interval of $(f_{r1}, f_s/6)$ when $f_{r1} < f_s/6$, where f_{r1} is the antiresonance frequency of the inverter-side inductor and capacitor of the LCL-filter as defined in (15). Also, when $f_{r1} > f_s/6$, $\text{Re}\{Y_2(j\omega)\}$ is negative in the interval of $(f_s/6, f_{r1})$. This has also been intuitively illustrated in Fig. 4(a). However, since $f_r > f_{r1}$ is always true in an inductive grid, so that f_r is always in the passive region of $Y_2(s)$ as long as $f_r > f_s/6$, regardless of $f_{r1} < f_s/6$ or $f_{r1} > f_s/6$. Thus, the system is stable, and vice versa. This is confirmed by the Nyquist curves in Fig. 4(b). Note that the stable range identified in this article via passivity theory is in line with the results obtained via the discrete-time transfer

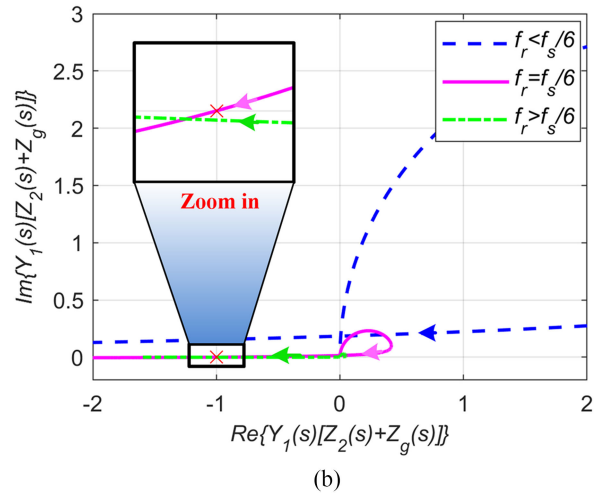
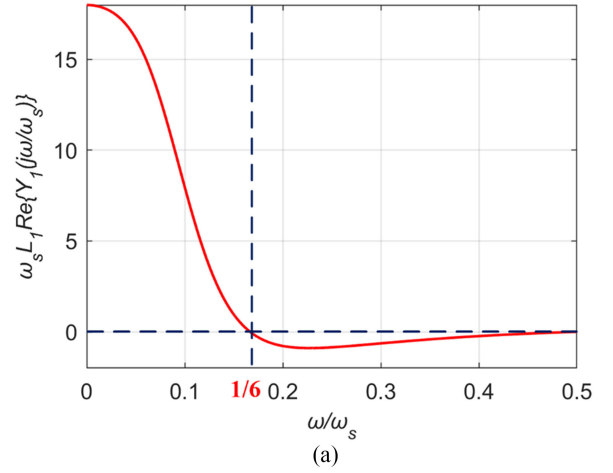


Fig. 3. Single-loop inverter-side current control. (a) Real part of the output admittance. (b) Nyquist curves for different system resonant frequencies.

function based method in [15]

$$f_{r1} = \frac{1}{2\pi} \sqrt{\frac{1}{L_1 C}}. \quad (15)$$

D. CCF-AD

For a sake of simplicity, the proportional CCF-AD with the gain, K_{ad} , is used for the passivity property analysis. Then, substituting $H_{ad}(s) = K_{ad}$ into (9), taking the real part of $Y_{ad}(s)$ yields

$$\text{Re}\{Y_{ad}(j\omega)\} = \frac{K_{ad} C \omega^2 \cos(\omega T_d)}{[\omega_c^2 + \omega^2 - 2\omega_c \omega \sin(\omega T_d)]L_1}. \quad (16)$$

It is obvious that $\text{Re}\{Y_{ad}(j\omega)\}$ has a similar property to $\text{Re}\{Y_1(j\omega)\}$, which also changes sign from positive to negative at $\omega = \omega_s/6$, as shown in Fig. 5.

1) *Inverter-Side Current Control Plus “CCF-AD”*: The total output admittance ($Y_{1ad}(s)$) equals to the summation of $Y_1(s)$ and $Y_{ad}(s)$, taking the real part of which yields

$$\text{Re}\{Y_{1ad}(j\omega)\} = \frac{(K_{ad} C \omega^2 + \omega_c) \cos(\omega T_d)}{[\omega_c^2 + \omega^2 - 2\omega_c \omega \sin(\omega T_d)]L_1}. \quad (17)$$

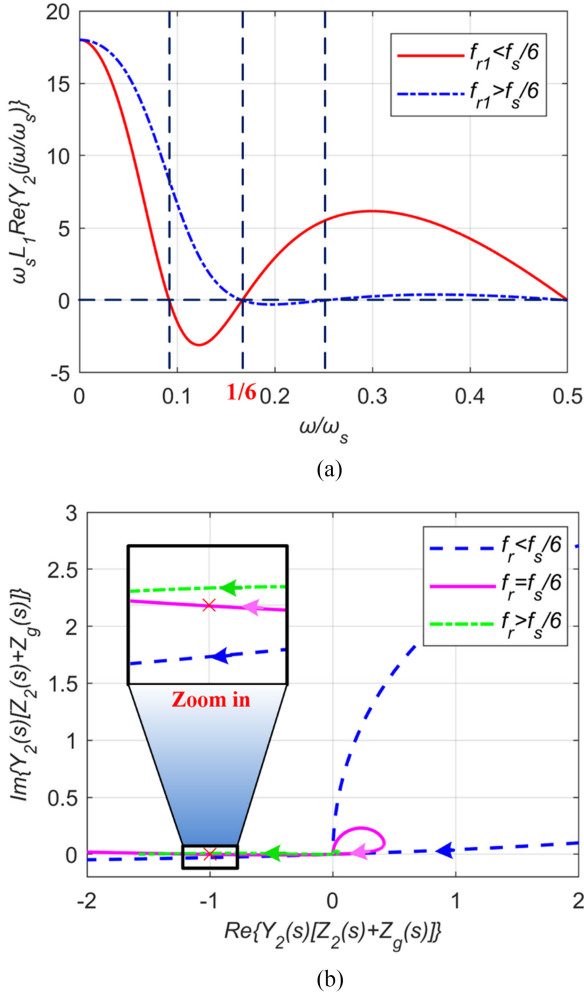


Fig. 4. Single-loop grid-side current control. (a) Real part of the output admittance. (b) Nyquist curves for different system resonant frequencies.

The sign change at $\omega = \omega_s/6$ caused by the cosine function in the numerator of (17) can be increased by selecting K_{ad} , such that the factor $(K_{ad}C\omega^2 + \omega_c)$ changes sign exactly when the cosine function does. This is achieved by letting

$$K_{ad} = -\frac{36\omega_c}{C\omega_s^2}. \quad (18)$$

With this selection, the nonnegative real part for all frequencies is verified in Fig. 5. Similar methodology has been applied in [30].

2) *Grid-Side Current Control Plus “CCF-AD”*: The total output admittance ($Y_{2ad}(s)$) equals to the summation of $Y_2(s)$ and $Y_{ad}(s)$, taking the real part of which yields

$$\text{Re}\{Y_{2ad}(j\omega)\} = \frac{[(K_{ad} - L_1\omega_c)C\omega^2 + \omega_c]\cos(\omega T_d)}{[\omega_c^2 + \omega^2 - 2\omega_c\omega\sin(\omega T_d)]L_1}. \quad (19)$$

The factor $[(K_{ad} - L_1\omega_c)C\omega^2 + \omega_c]$ in the numerator of (19) will change sign exactly when the cosine function does if K_{ad} is chosen as follows:

$$K_{ad} = \omega_c L_1 - \frac{36\omega_c}{C\omega_s^2}. \quad (20)$$

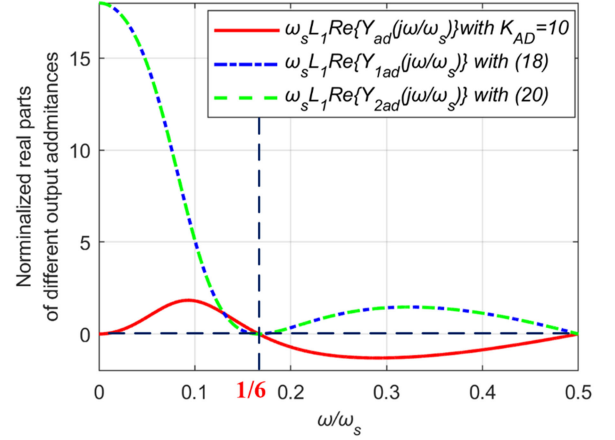


Fig. 5. Real parts for output admittance of CCF-AD with $H_{ad}(s) = K_{ad}$ and its applications in inverter-side or grid-side current control.

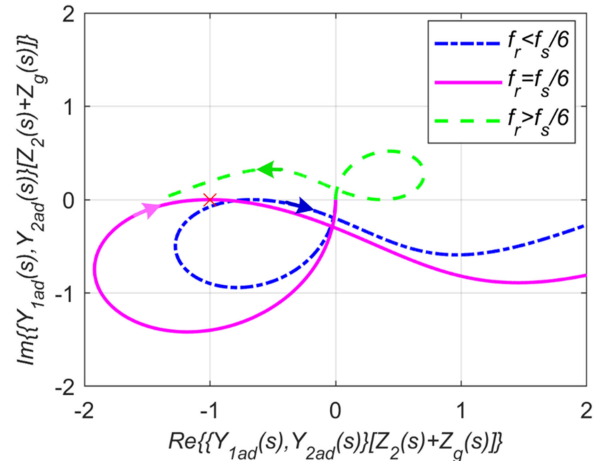


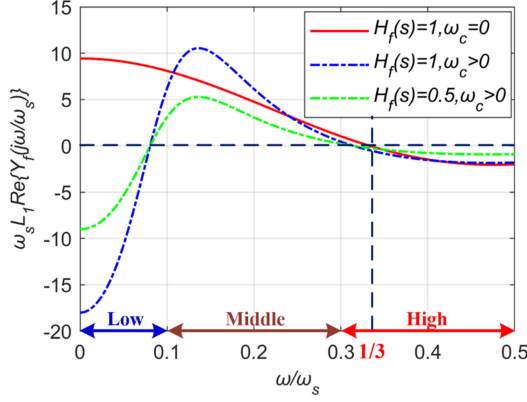
Fig. 6. Nyquist curves for either inverter-side or grid-side current control plus “CCF-AD” under different system resonant frequencies.

With this choice, $\text{Re}\{Y_{2ad}(j\omega)\}$ will be exactly equal to $\text{Re}\{Y_{1ad}(j\omega)\}$ designed with (18), so the curve of $\text{Re}\{Y_{2ad}(j\omega)\}$ will overlap with the one of $\text{Re}\{Y_{1ad}(j\omega)\}$ as shown in Fig. 5. The nonnegative real part for all frequencies is also realized for the grid-side current control by using CCF-AD.

Substituting (11), (18), and (20) into (9), not only the real part of the admittance but also the whole equivalent admittance is equal to each other for both inverter-side and grid-side current control plus CCF-AD. Thanks to the nonnegative real part for all frequencies, the system stability will be immune to the system resonant frequency. This is intuitively verified by Nyquist curves shown in Fig. 6. Note that the system is critical stable when $f_r = f_s/6$, due to the zero real part of the output admittance at $f_s/6$.

E. CVF-AD

For the sake of simplicity, the proportional CVF-AD with the gain, K_f , is used for the passivity property analysis. Then, substituting $H_f(s) = K_f$ into (9), taking the real part of $Y_f(s)$

Fig. 7. Real part for output admittance of CVF-AD with $H_f(s) = K_f$.

yields

$$\text{Re}\{Y_f(j\omega)\} = \frac{K_f[\omega \sin(\omega T_d) - \omega_c]}{[\omega_c^2 + \omega^2 - 2\omega_c\omega \sin(\omega T_d)]L_1}. \quad (21)$$

When ignoring the influence of the current control loop, i.e., $\omega_c = 0$, $\text{Re}\{Y_f(j\omega)\}$ changes sign from positive to negative at $\omega = \omega_s/3$. As ω_c increases, the passive region will shrink toward the center, as shown in Fig. 7.

It can be inferred from Figs. 3, 4, and 7 that CVF can roughly extend the frequency region of passive output admittance (in other words, stable region) of the single-loop inverter-side or grid-side current control to $f_s/3$, in line with the results found in [23] and [24]. Due to the length limitation of this article, how to optimize the passivity of the single-loop inverter-side or grid-side current control using only CVF-AD will not be discussed in this article, which would be the future work.

Although CCF-AD has turned the real part of the output admittance to be nonnegative for all frequencies for both inverter-side and grid-side current-control, the passivity is very weak around $f_s/6$. This can be exactly enhanced by the CVF-AD, but along with the side effect of passivity losses at both low and high frequencies. One simple and effective way to alleviate this side effect is to reduce the value of K_f , but it still suffers from the problem of incomplete elimination of nonpassivity at high frequencies. In order to completely eliminate the side effect induced by the CVF-AD at high frequencies and ensure all frequency passivity, a first-order finite-impulse response (FIR) low-pass filter (LPF) in the form of $G_{\text{LPF}}(s) = 1 - a + ae^{-sT_s}$ ($0 < a < 1$) is used due to its simplicity in digital implementation. Then, $H_f(s) = K_f$ is replaced by $H_f(s) = K_f G_{\text{LPF}}(s)$ in practice.

Fig. 8 shows the real part of the output admittance for either inverter-side or grid-side current control plus “CCF-AD” and “CVF-AD” with different values of a and K_f . In Fig. 8(a), with the increase of a , the passivity increases at high frequencies but decreases at middle frequencies. $a = 0.6$ would be a reasonable choice for achieving a compromise passivity between middle and high frequencies. In Fig. 8(b), with the increase of K_f , the passivity decreases at low frequencies but increases at middle frequencies. $K_f = 0.4$ would be a satisfactory tradeoff choice.

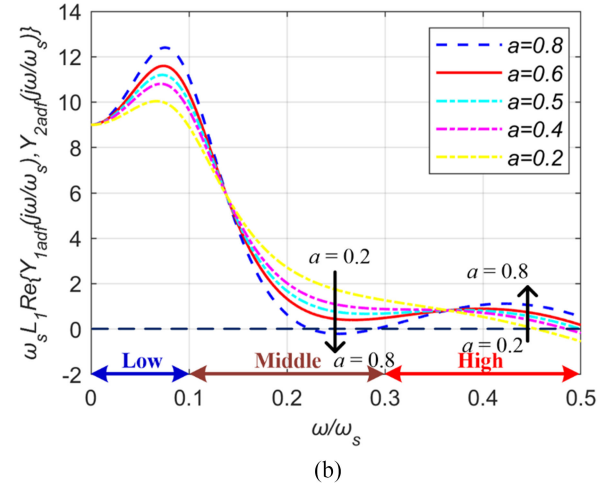
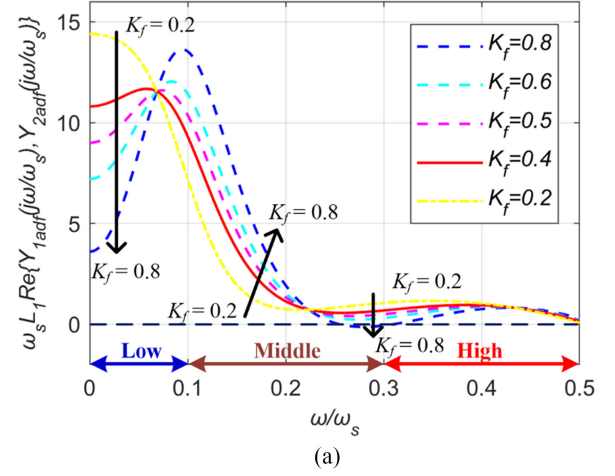


Fig. 8. Real part of the output admittance for either inverter-side or grid-side current control plus “CCF-AD with $H_{ad}(s) = K_{ad}$ ” and “CVF-AD with $H_f(s) = K_f G_{\text{LPF}}(s)$.” (a) $K_f = 0.5$ with different value of a . (b) $a = 0.6$ with different value of K_f .

IV. UNIFIED ANALYTICAL CONTROLLER PARAMETERS DESIGN METHOD

A. Parameters Selection of Current Controller $G_c(s)$

According to (7), both the inverter-side and grid-side current control share the same first-order system with a pure delay element, whereby the inner loop current controller parameters can be designed according to the following equivalent open-loop transfer function:

$$T_o(s) = G_c(s)G_d(s)/L_1s. \quad (22)$$

The phase angle of this open-loop transfer function at the crossover angular frequency ω_c is given by

$$\begin{aligned} \angle T_o(j\omega_c) &= \angle \left\{ G_c(j\omega_c)G_d(j\omega_c) \frac{1}{sL_1} \right\} \\ &= \angle \left\{ \left(K_p + K_I \frac{j\omega_c \cos\phi_1 - \omega_1 \sin\phi_1}{\omega_1^2 + \omega_c \omega_{rc} - \omega_c^2} \right) \right. \\ &\quad \left. \times e^{-j\omega_c T_d} \frac{1}{j\omega_c L_1} \right\} \end{aligned}$$

$$\begin{aligned}
\omega_c \gg \omega_1 \gg \omega_{rc} &\approx \angle \left\{ \left(K_p - K_I \frac{j \cos \phi_1}{\omega_c} \right) e^{-j \omega_c T_d} \frac{1}{j \omega_c L_1} \right\} \\
&= -\arctan \frac{K_I \cos \phi_1}{K_p \omega_c} - \omega_c T_d - \frac{\pi}{2} \\
&= -\pi + \varphi_m
\end{aligned} \tag{23}$$

where ϕ_m is the desired phase margin (PM) of the open-loop transfer function. The crossover frequency ω_c can be rewritten as

$$\omega_c = \frac{\pi/2 - \varphi_m - \varphi_0}{T_d} \tag{24}$$

where $\phi_0 = \arctan(K_I \cos \phi_1 / K_p \omega_c)$. It can be observed from (24) that the value of ω_c for a given ϕ_m is inversely proportional to the value of ϕ_0 . Thus, ϕ_0 can be designed to be sufficiently small to achieve a large value of ω_c [32]. When ϕ_0 is small enough, $\phi_0 \approx K_I \cos \phi_1 / K_p \omega_c$. Then, the gain of the resonant controller can be easily calculated as

$$K_I = K_p \omega_c \varphi_0 / \cos \phi_1 \tag{25}$$

in which the proportional gain K_p can be found by setting the open-loop gain at the value of ω_c to unity using (22), and the phase compensation angle φ_1 is set to cancel the computation and PWM delays [33], which, respectively, give

$$K_p \approx \omega_c L_1 \tag{26}$$

$$\phi_1 = \omega_1 T_d. \tag{27}$$

B. Parameters Selection of CCF-AD

According to the analysis in Section III-D, when using CCF-AD, both the inverter-side and grid-side current control can obtain the nonnegative real part of the output admittance from 0 to Nyquist frequency. The capacitor current feedback gains can be directly calculated using (18) or (20) accordingly.

C. Parameters Selection of CVF-AD

As analyzed in Section III-E, CVF-AD can help to enhance the passivity of the output admittance at middle frequencies. However, the performance is affected by CVF-AD gain. Meanwhile, the capacitor voltage can often be used for inrush current suppression at start-up, grid synchronization and grid voltage disturbance rejection [34]. In practice, the CVF-AD controller $H_f(s)$ consists of the aforesaid first-order FIR-LPF $G_{LPF}(s)$ scaled by K_f and a bandpass filter (BPF) as follows:

$$\begin{aligned}
H_f(s) &= K_f G_{LPF}(s) + K_{fb} G_{BPF}(s) \\
&= K_f \underbrace{(1 - a + a e^{-s T_s})}_{G_{LPF}(s)} + K_{fb} \omega_{bc} \underbrace{\frac{s \cos \phi_b - \omega_1 \sin \phi_b}{s^2 + \omega_{bc} s + \omega_1^2}}_{G_{BPF}(s)}
\end{aligned} \tag{28}$$

where K_{fb} , ω_{bc} , and φ_b are the gain, cut-off frequency, and compensation angle of BPF, respectively. The coefficients a and K_f

in $H_f(s)$ have already been defined and chosen in Section III-E. The rest three parameters in $H_f(s)$ can be designed as follows.

- 1) ω_{bc} is set to $0.1 \omega_1$ to achieve a compromise between frequency selectivity and dynamic response.
- 2) When $H_f(j \omega_1) \approx e^{j \omega_1 T_d}$, the total time delay at the fundamental frequency can be well compensated. This is achieved by letting

$$\phi_b = \arctan \frac{\sin(\omega_1 T_d)}{\cos(\omega_1 T_d) - K_f} \tag{29}$$

and

$$K_{fb} = \frac{\sin(\omega_1 T_d)}{\sin \phi_b} \approx 1 - K_f. \tag{30}$$

D. Digital Implementation

Issues that are related to the digital implementation of the control law are considered in this article. There are two controllers that need to be discretized into z-domain, i.e., $G_c(s)$ and $H_f(s)$. The FIR-LPF $G_{LPF}(s)$ in $H_f(s)$ can be directly rewritten in z-domain as $G_{LPF}(z) = 1 - a + a z^{-1}$ without any digital approximation accuracy loss. The resonators in $G_c(s)$ and the $G_{BPF}(s)$ in $H_f(s)$ can be constructed with two s-domain integrators which, and then discretized into z-domain using the forward Euler and backward Euler method [35], [36] to achieve high digital approximation accuracy. Taking into consideration that both resonators in $G_c(s)$ and $G_{BPF}(s)$ in $H_f(s)$ only contribute to the component around the fundamental frequency, so that the influence of the digital discretization as well as the resonators themselves on the passivity of the output admittance, viz., the damping region, can be ignored.

V. EXPERIMENTAL VERIFICATIONS

For validating the correctness of the theoretical analysis and the effectiveness of the proposed controller parameter design method, an experimental setup is built up in the laboratory, as shown in Fig. 9. The setup consists of four three-phase inverters with LCL filters, one of them is performed as the grid-connected inverter under test. The control algorithm is implemented in the dSPACE 1005 platform for real-time control. The power stage parameters are listed in Table I for the controller parameters design.

According to the knowledge of the control theory [37], the damping of the closed-loop system is proportional to the open-loop PM ϕ_m which is hence set to $11\pi/36$ rad (55°) to obtain sufficient damping for the inner loop. The coefficient ϕ_0 is chosen as 0.1 rad. Then, substituting the values of ϕ_m , ϕ_0 , and T_d ($1.5/f_s$) into (24), ω_c can be calculated, that is, $\omega_c = 0.0545 \omega_s$. The Bode plot of the open-loop transfer function of the inner loop is drawn in Fig. 10, where the PM and crossover angular frequency are almost as expected. All controller parameters can directly be calculated using (18), (20), (25), (26), (27), (29), and (30). The results are presented in Table II.

Since the calculation of controller parameters depends on some power circuit parameters, i.e., inverter-side inductance in (18), (20), and (26), filter capacitance in (20). The influence of

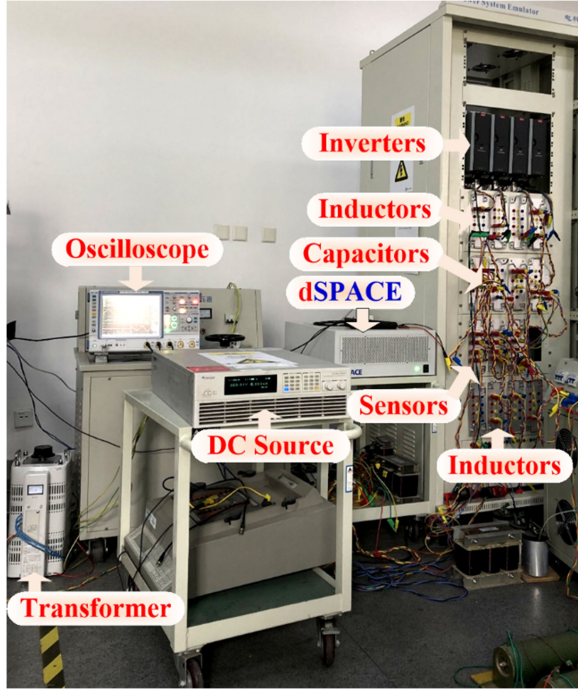


Fig. 9. Hardware picture for the experimental setup.

TABLE I
POWER STAGE PARAMETERS

	Symbol	Description	Value
Inverter	V_{dc}	Input DC voltage	350 V
	L_1	Inverter-side inductor	1.8 mH
	C	Capacitor of LCL-filter	9.8 μ F
	L_2	Grid-side inductor	0.4 mH
	f_s	Sampling and switching frequency	10 kHz
	ω_s	Sampling angular frequency	62832 rad/s
Grid	V_g	Grid voltage (Phase-to-ground RMS Voltage)	110 V
	f_g	Frequency	50 Hz
	L_g	Grid leakage inductor	0.2mH, 1.6mH, 6.6mH
	f_{r1}	System resonant frequency	$0.24f_s$, $0.167f_s$, $0.13f_s$
	C_p	PFC capacitor	15 μ F (0.15 P.U.)

variations of those power circuit parameters on the passivity of the inverter output admittance is analyzed. Fig. 11 shows the real part of the output admittance [Ω^{-1}] with the aforedesigned inverter-side and grid-side current control parameters in variations of the inverter-side inductance or filter capacitance. As seen from Fig. 11, all frequencies passive output admittance can be ensured if the inverter-side inductor and filter capacitor have tolerances less than 32% and 26%, respectively. Note that this is a pessimistic value since a large tolerance is reasonably expected if the controller parameters are designed according to the minimum values of power circuit parameters.

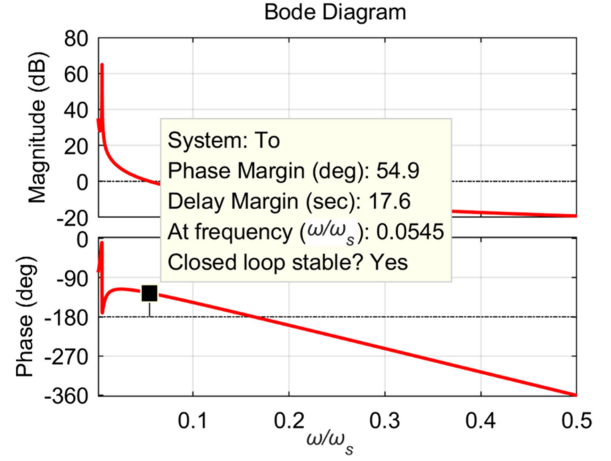


Fig. 10. Bode plot of the open-loop transfer function of the inner loop.

TABLE II
CONTROLLER PARAMETERS

	Symbol	Description	Value
Current Controller	K_p	Proportional gain	6.13 Ω
	K_i	Integral gain	2088 Ω/s
	ω_1	Resonant angular frequency	314 rad/s
	ω_{rc}	Cut-off angular frequency	0.003 rad/s
	ϕ_1	Compensation angle	0.07 rad
CCF-AD	K_{ad}	CCF gain for inverter-side current control	-3.17 Ω
	K_{ad}	CCF gain for grid-side current control	2.96 Ω
CVF-AD	K_f	FIR-LPF gain for CVF	0.4
	K_{fb}	BPF gain for CVF	0.6
	a	Coefficient of FIR-LPF	0.6
	ω_{bc}	Cut-off angular frequency	31.4 rad/s
	ϕ_b	Compensation angle	0.08 rad

A. Tests When the System Resonant Frequency f_r Beyond $f_s/6$

Experiments in the scenario of the system resonant frequency beyond $f_s/6$ are first performed, i.e., $L_g = 0.2$ mH and $f_r = 0.24 f_s$. The experimental waveforms for the inverter-side and grid-side current control are given in Figs. 12 and 13, respectively.

As seen from Fig. 12(a), when CCF-AD and CVF-AD are disabled, the inverter-side current controlled inverter enters instability (about 2.4 kHz oscillation), which validates the analysis in Section III-D, that is, the passivity of the output admittance of the inverter-side current controlled inverter at frequencies beyond $f_s/6$ is provided by either CCF-AD or CVF-AD. And Fig. 12(b) gives dynamic experimental waveforms when CCF-AD and CVF-AD are enabled. The experimental results verify the effectiveness of the proposed method in guaranteeing the good dynamic performance of the inverter-side current control in this scenario.

Since the single-loop grid-side current control can keep stability in the presence of the system resonant frequency beyond

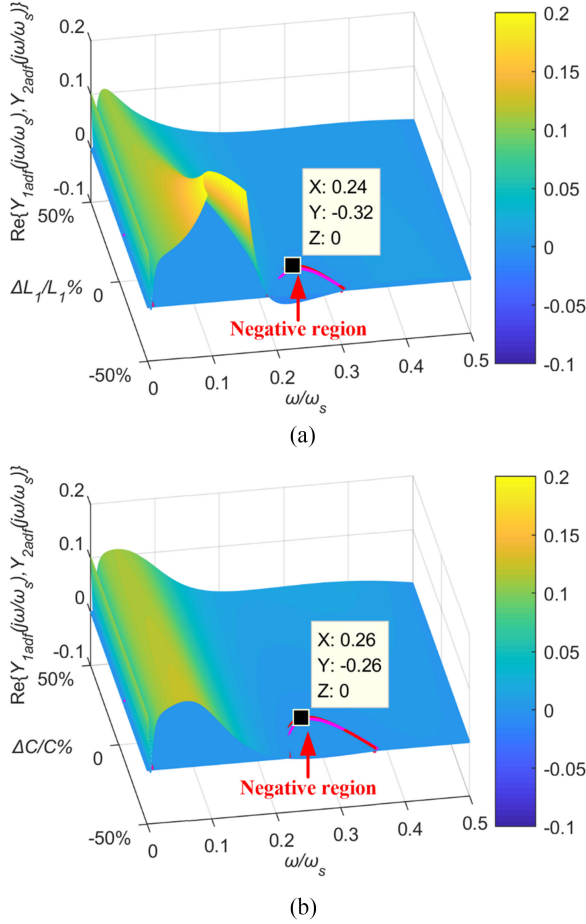


Fig. 11. Real part of the output admittance [Ω^{-1}] with aforedesigned inverter-side and grid-side current control parameters in variations of (a) inverter-side inductance or (b) filter capacitance.

$f_s/6$, Fig. 13 only gives the dynamic experimental waveform of the step change of reference current to confirm the theoretical expectation.

B. Tests When the System Resonant Frequency f_r Near $f_s/6$

Figs. 14 and 15, respectively, give dynamic experimental waveforms of the step change of reference current for the inverter-side and grid-side current control when the system resonant frequency is close to $f_s/6$, i.e., $L_g = 1.6$ mH and $f_r = 0.167 f_s$.

As shown in Figs. 14(b) and 15(b), when CVF-AD is disabled, there is obvious oscillation in output current during the step change of reference current process due to the poor passivity of the output admittance at the system resonant frequency is poor when only CCF-AD is enabled. It takes about half a fundamental period to reach the steady state.

By contrary, when CVF-AD is enabled, the passivity of the output admittance at frequencies around $f_s/6$ is enhanced; thus, oscillation of output current is prevented as shown in Figs. 14(a) and 15(a). It confirms the theoretical result in Section III-E, i.e., CVF-AD is able to increase the passivity of the output admittance at frequencies around $f_s/6$.

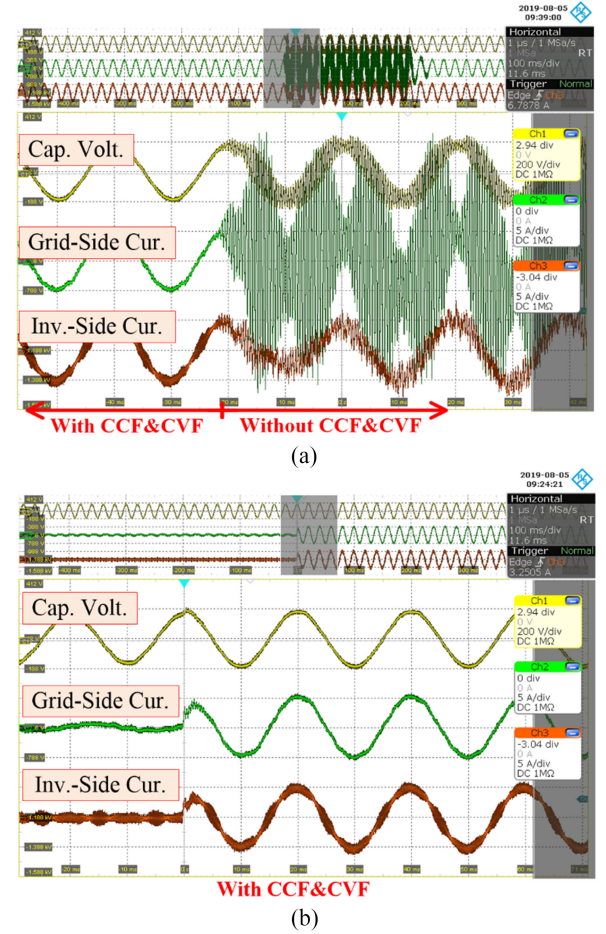


Fig. 12. Experimental waveforms of the inverter-side current control when the system resonant frequency beyond $f_s/6$. (a) Steady state. (b) Step change of reference current.

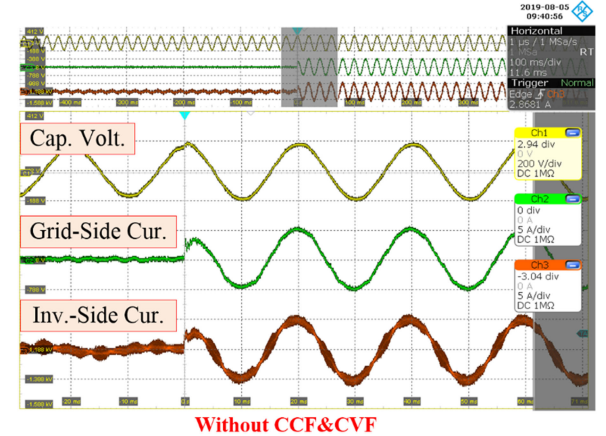
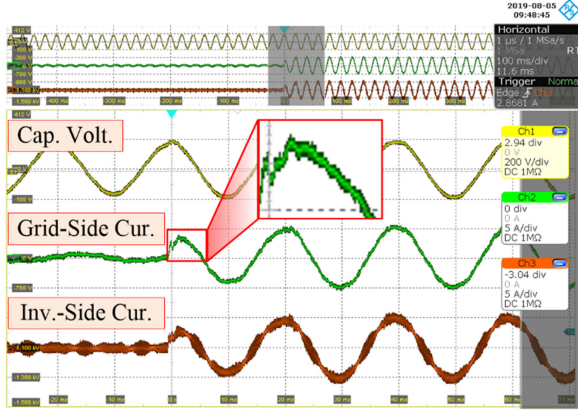


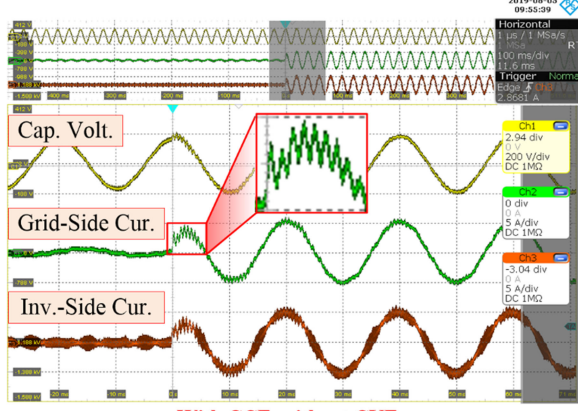
Fig. 13. Dynamic experimental waveform of the step change of reference current for the grid-side current control when the system resonant frequency beyond $f_s/6$.

C. Tests When the System Resonant Frequency f_r Below $f_s/6$

Experiments are performed when the system resonant frequency is below $f_s/6$, i.e., $L_g = 6.6$ mH and $f_r = 0.13 f_s$. The experimental waveforms are given in Figs. 16 and 17.



(a)



(b)

Fig. 14. Dynamic experimental waveforms of the step change of reference current for the inverter-side current control (a) with or (b) without CVF-AD when the system resonant frequency is close to $f_s/6$.

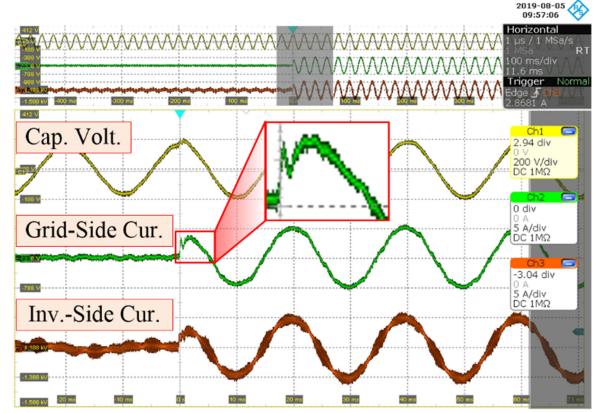
Since the single-loop inverter-side current control can keep stability in this scenario, Fig. 16 only gives the dynamic experimental waveforms to confirm the theoretical expectation.

In Fig. 17(a), when CCF-AD and CVF-AD are disabled, the grid-side current controlled inverter will enter instability (about 1.3-kHz oscillation) and even overcurrent trip, due to the loss of the passivity of the output admittance at frequencies below $f_s/6$ provided by either CCF-AD or CVF-AD. Fig. 17(b) gives dynamic experimental waveforms of the grid-side current control with CCF-AD and CVF-AD to show the effectiveness of the proposed in ensuring good dynamic performance in this scenario.

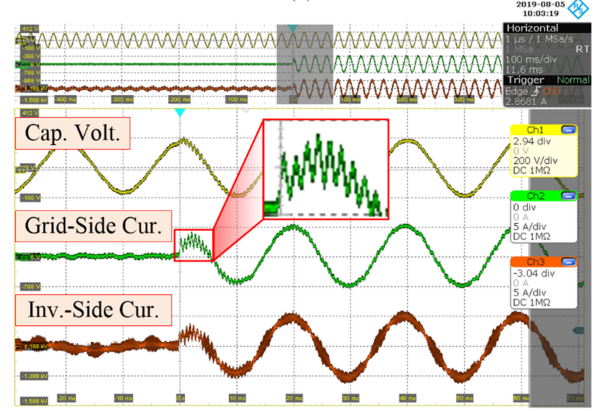
D. Tests in the Capacitive Grid Condition

Finally, experiments are performed when a PFC capacitor is connected to PCC. According to Fig. 2, seen from PCC, the output admittance can be redefined as $Y_t(s) = Y(s)/[Z_2(s) + Y(s)]$, and the grid impedance may be $Z_{g2}(s) = L_g s / (L_g C_p s^2 + 1)$ or $Z_{g1}(s) = L_g s$ when the PFC capacitor is or is not connected.

As verified by the above experimental results, single-loop inverter-side or grid-side controlled inverter can keep stability

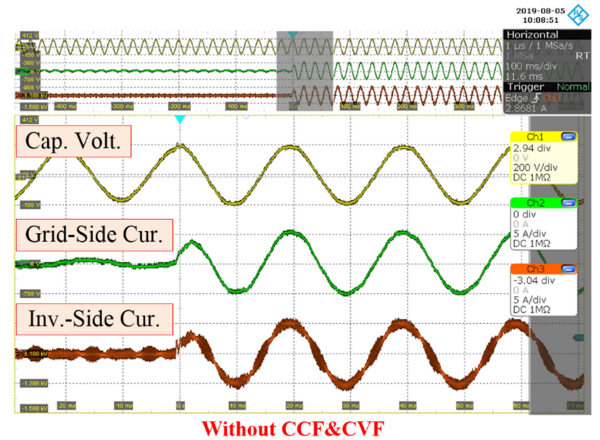


(a)



(b)

Fig. 15. Dynamic experimental waveforms of the step change of reference current for the grid-side current control (a) with or (b) without CVF-AD when the system resonant frequency is close to $f_s/6$.



Without CCF&CVF

Fig. 16. Dynamic experimental waveforms of the step change of reference current for the inverter-side current control under the system resonant frequency below $f_s/6$.

if the system resonant frequency is below or beyond $f_s/6$ in case the grid is inductive. However, this stability condition may be challenged by the PFC capacitor. Fig. 18 gives theoretical expectations in terms of Nyquist curves of $Z_g(s)Y_t(s)$. Since there is a pair of right-half plane poles in $Z_g(s)Y_t(s)$, according

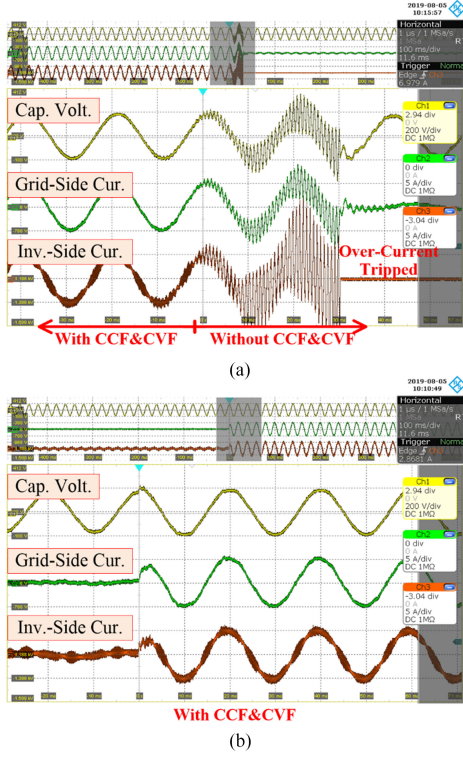


Fig. 17. Experimental waveforms of the grid-side current control when the system resonant frequency is below $f_s/6$. (a) Steady state. (b) Step change of reference current.

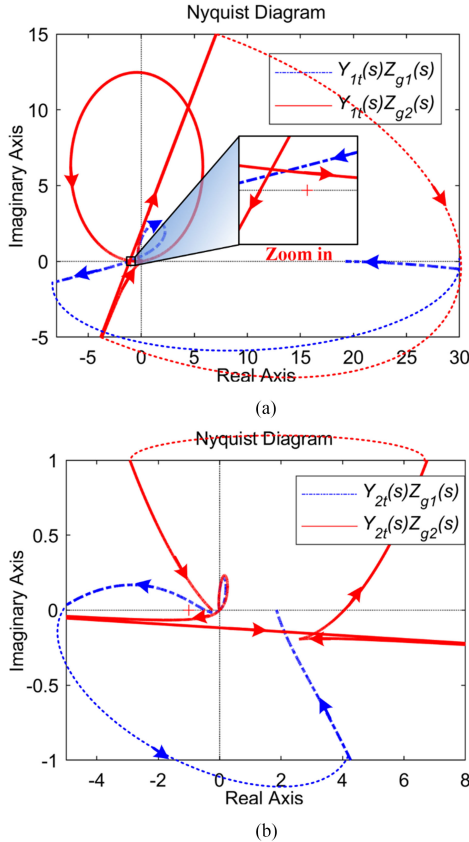


Fig. 18. Nyquist curves of $Z_g(s)Y_t(s)$ for the single loop. (a) Inverter side or (b) grid-side current control when the PFC capacitor is or is not connected to PCC.

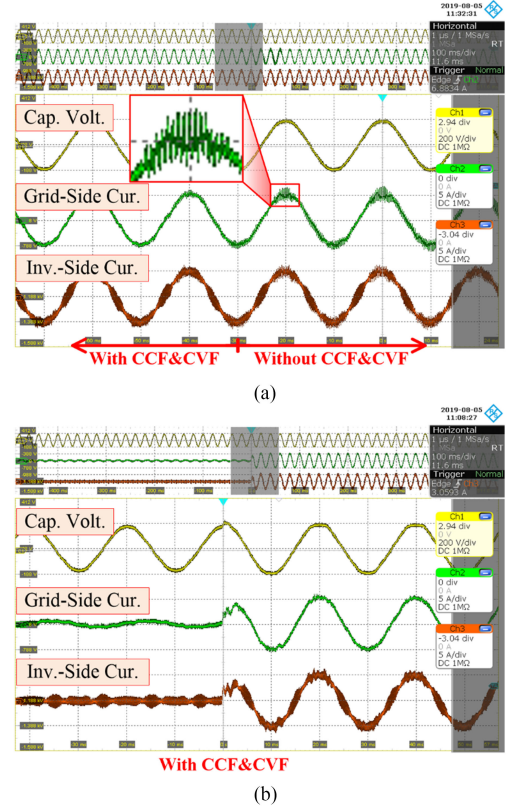


Fig. 19. Experimental waveforms of the inverter-side current control when the PFC capacitor is connected to PCC ($L_g = 5.6$ mH, $C_p = 15$ μ F). (a) Steady state. (b) Step change of reference current.

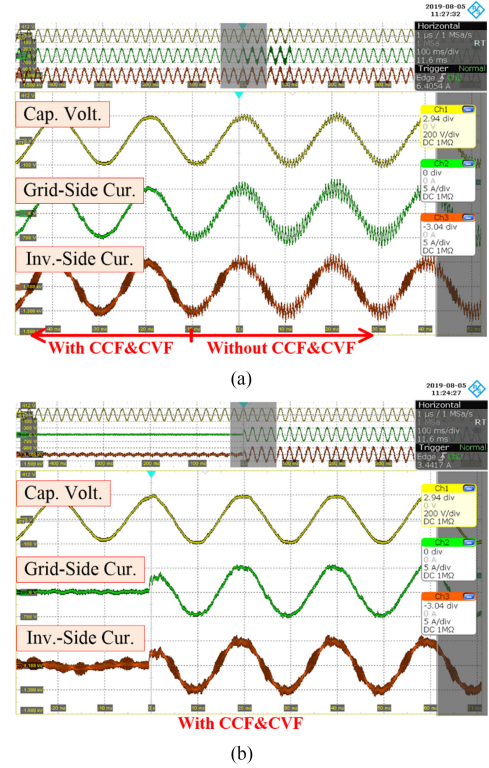


Fig. 20. Experimental waveforms of the grid-side current control when the PFC capacitor is connected to PCC ($L_g = 0.6$ mH, $C_p = 15$ μ F). (a) Steady state. (b) Step change of reference current.

to the Nyquist criterion, the system is stable if the critical point $(-1, 0)$ is encircled by the Nyquist curve [37], and vice versa. As shown in Fig. 18, the Nyquist curves change from encirclement of the critical point to non-encirclement of the critical point when the PFC capacitor is connected, so that the system enters instability from stability condition. Note that L_g is 5.6 and 0.6 mH in Fig. 18(a) and (b), respectively.

Fig. 19 gives steady-state and dynamic experimental waveforms for the inverter-side current control when the PFC capacitor is connected to PCC.

As shown in Fig. 19(a), when both CCF-AD and CVF-AD are disabled, there is very high-frequency oscillation in the output current. Fig. 19(b) shows the dynamic experimental waveforms when CCF-AD and CVF-AD are enabled. The experimental results confirm the effectiveness of the proposed CCF-AD and CVF-AD in damping the oscillation and ensuring good dynamic performance.

Similar experimental results are obtained for the grid-side current control as shown in Fig. 20. The difference is that, when both CCF-AD and CVF-AD are disabled, the oscillation frequency is lower and the oscillation is more severe. The reason for the difference would lie in the different passivity of the output admittance for the single-loop inverter-side or grid-side current control as shown in Figs. 3 and 4.

VI. CONCLUSION

This article first builds a general admittance model to facilitate the passivity-based stability assessment and controller parameter optimization for both inverter-side and grid-side current-controlled LCL-type inverters. Then, an appropriate combination scheme of subadmittance as well as a simple unified analytical controller design method is proposed to obtain passivity in all frequencies for both inverter-side and grid-side current control. With the proposed design method, it can not only ensure the stable operation of the LCL-type inverter regardless of grid impedance (either inductive or capacitive) but also avoid the conventional design constraints of an LCL filter. In addition, the designed controller parameters are very robust against power circuit parameter variation, and the proposed control scheme is simple in digital implementation.

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